

Quick-Start Keyboard Control in the Virtual Machine

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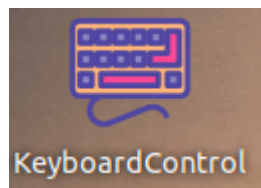
1. Course Content
2. Demonstration Case
3. Key Bindings
 - 3.1 Chassis Movement Control
 - 3.2 Speed Adjustment
 - 3.3 Gimbal (PTZ) Control

1. Course Content

- Use the desktop Keyboard Control icon to quickly start controlling the robot with the keyboard

2. Demonstration Case

- Double-click the **KeyboardControl** application icon on the desktop to start the keyboard control feature



- Click the **Keyboard UI** terminal with the mouse and follow the terminal prompts; use the keyboard to control the robot chassis movement and the gimbal servo angles

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Keyboard UI

For Holonomic mode (strafing), hold down the shift key:
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  U   I   O
  J   K   L
  M   <   >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

PTZ Control:
  1/2 : PTZ x axis +/- (range -90 ~ 90)
  3/4 : PTZ y axis +/- (range -90 ~ 20)
  0    : reset PTZ to center (x:0, y:-45)

CTRL-C to quit

currently:      speed 0.25      turn 1.0
```

3. Key Bindings

3.1 Chassis Movement Control

Key	Function
 u	Move forward-left (forward + turn left)
 i	Move forward
 o	Move forward-right (forward + turn right)
 j	Turn left in place
 k	Stop
 l	Turn right in place
 m	Move backward-left (backward + turn left)
 ,	Move backward
 .	Move backward-right (backward + turn right)

3.2 Speed Adjustment

Key	Function
q	Increase both the max linear speed and angular speed by 10%
z	Decrease both the max linear speed and angular speed by 10%
w	Increase only the linear speed by 10%
x	Decrease only the linear speed by 10%
e	Increase only the angular speed by 10%
c	Decrease only the angular speed by 10%

3.3 Gimbal (PTZ) Control

Key	Function
1	Increase the gimbal X-axis angle (range -90 ~ 90)
2	Decrease the gimbal X-axis angle (range -90 ~ 90)
3	Increase the gimbal Y-axis angle (range -90 ~ 20)
4	Decrease the gimbal Y-axis angle (range -90 ~ 20)
0	Reset the gimbal to the center position (x:0, y:-45)

Tip: During keyboard control, keep the **Keyboard UI** terminal window focused; otherwise, the keys will not take effect.

[!NOTE]

To stop keyboard control, press ctrl+c in the opened terminal to end the process and close the terminal window